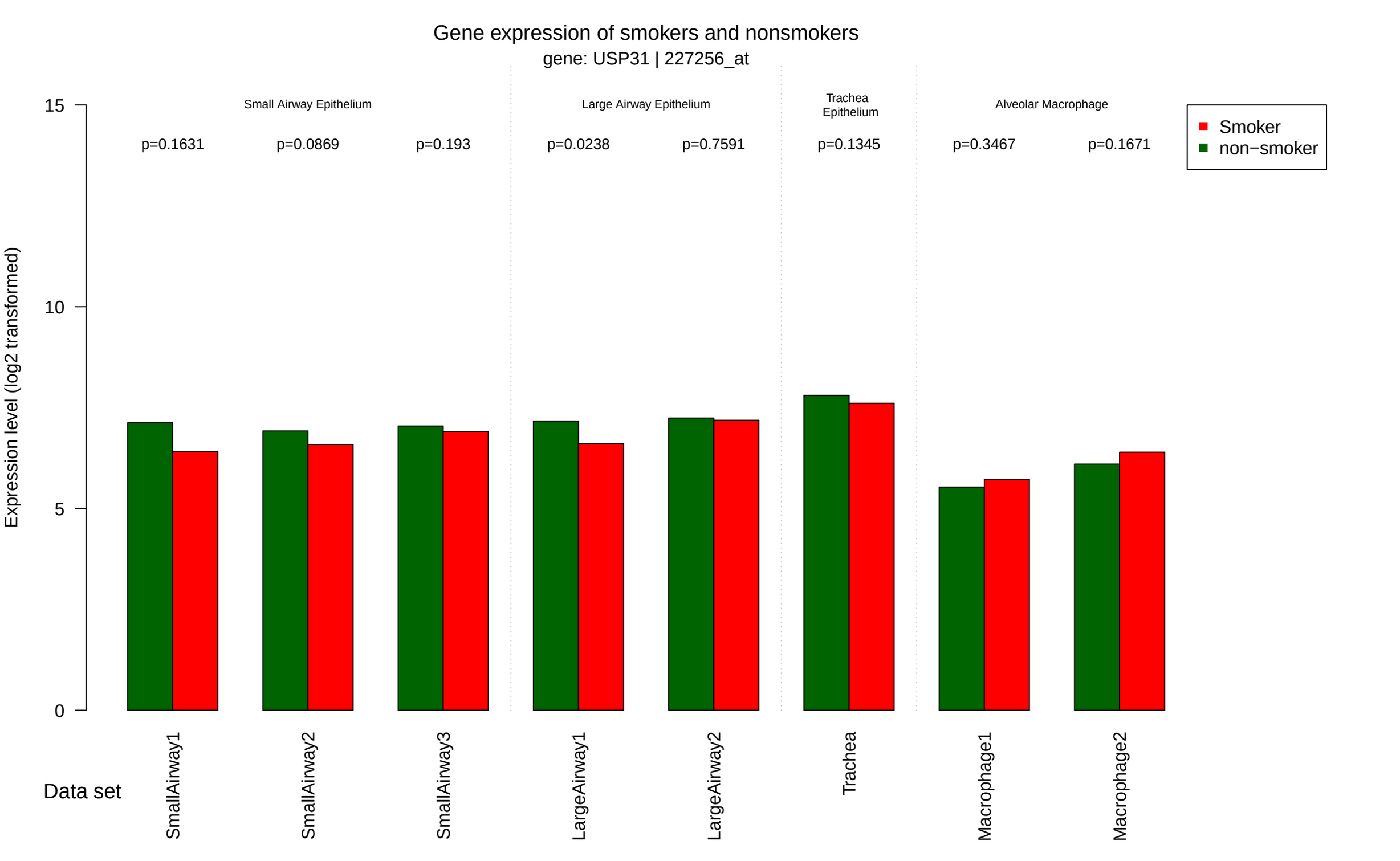
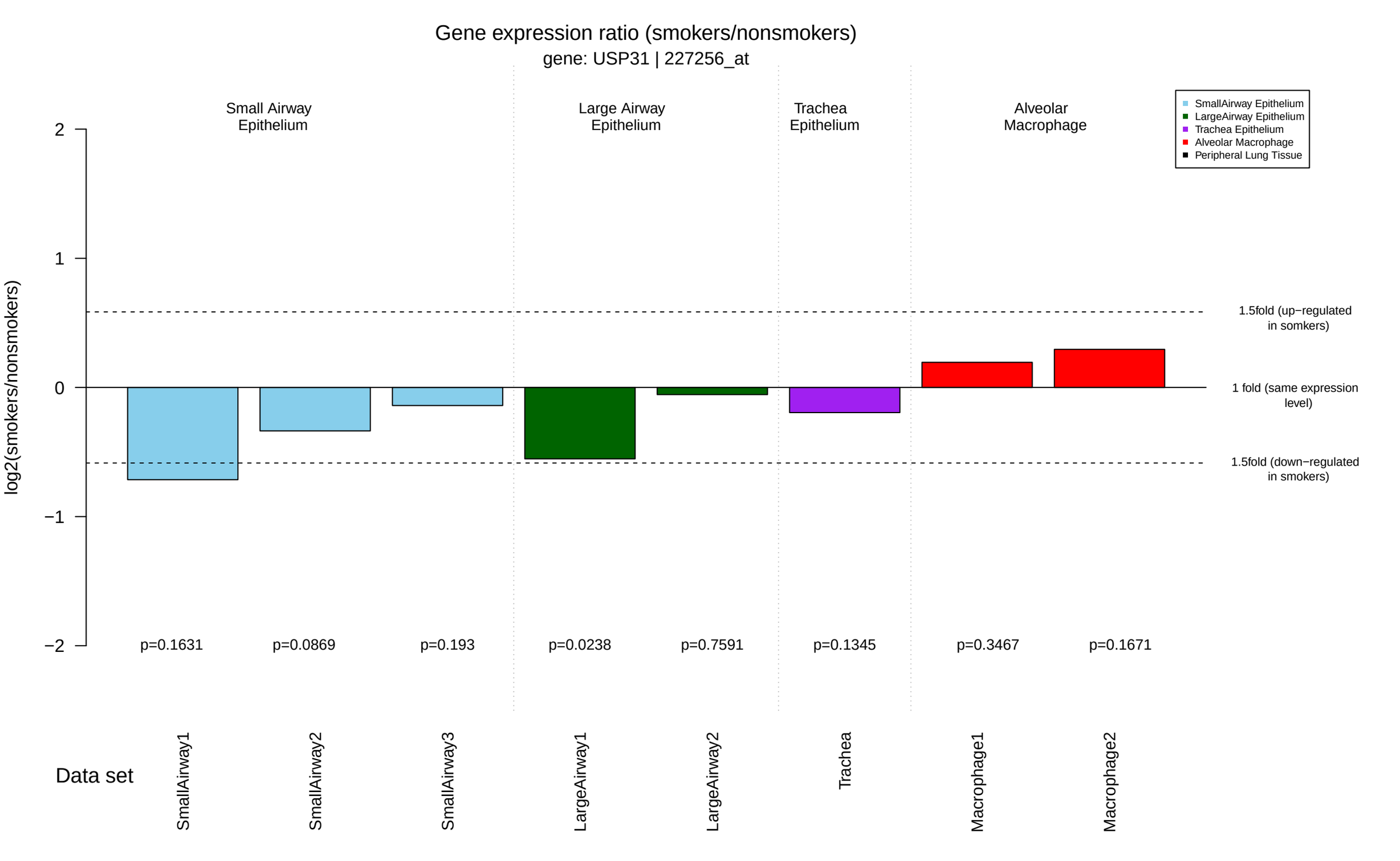
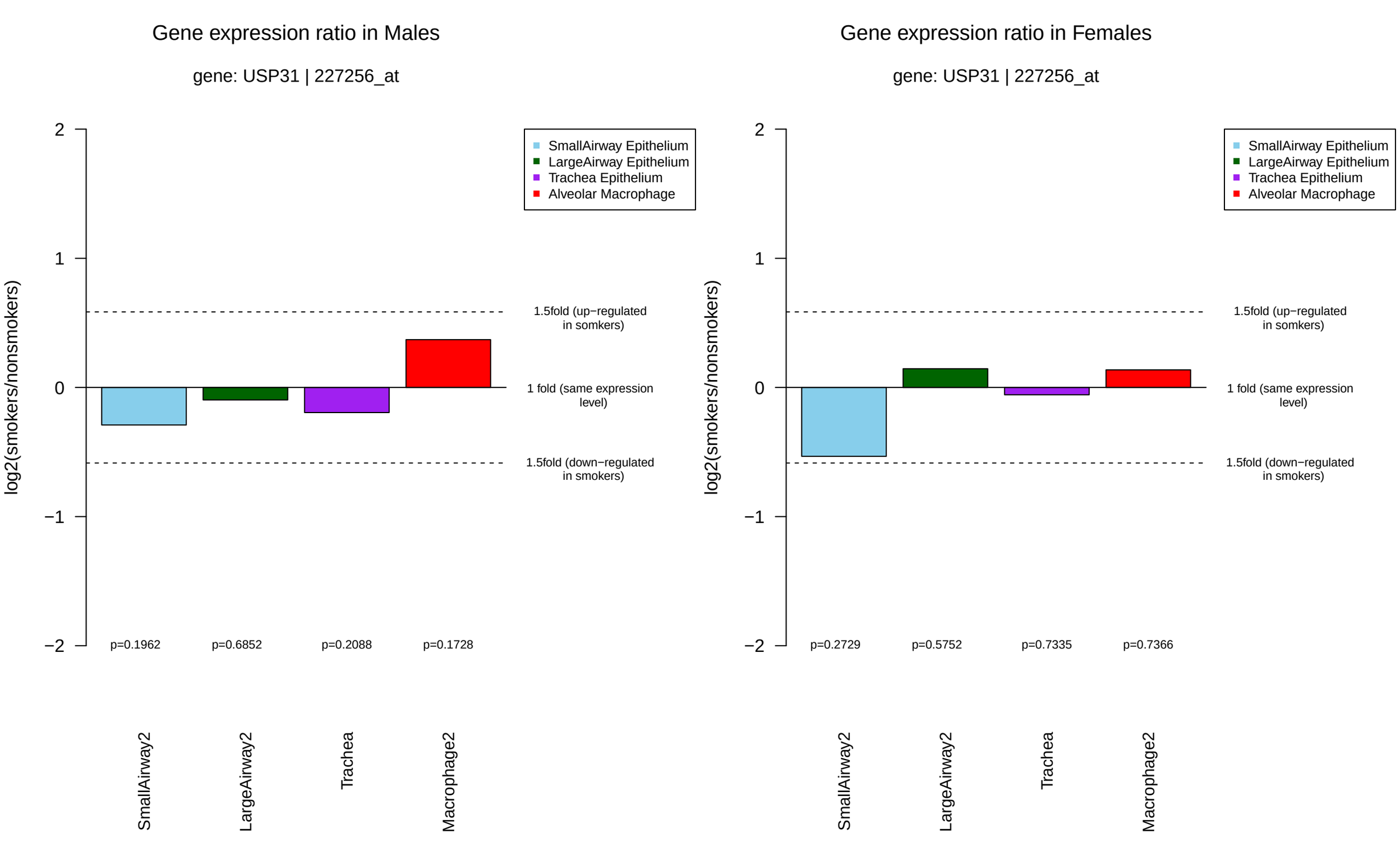
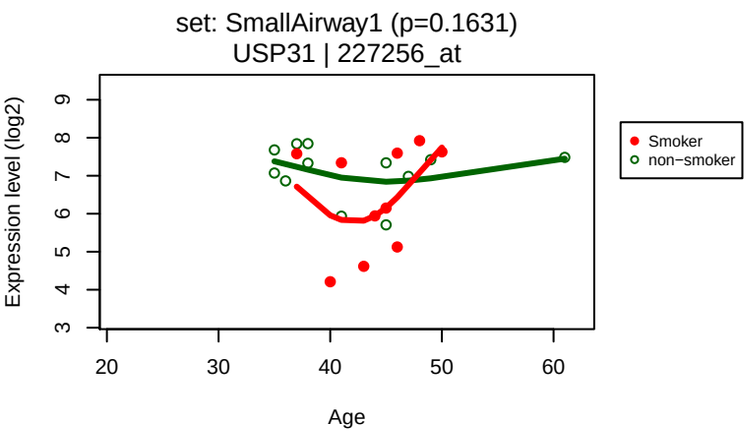
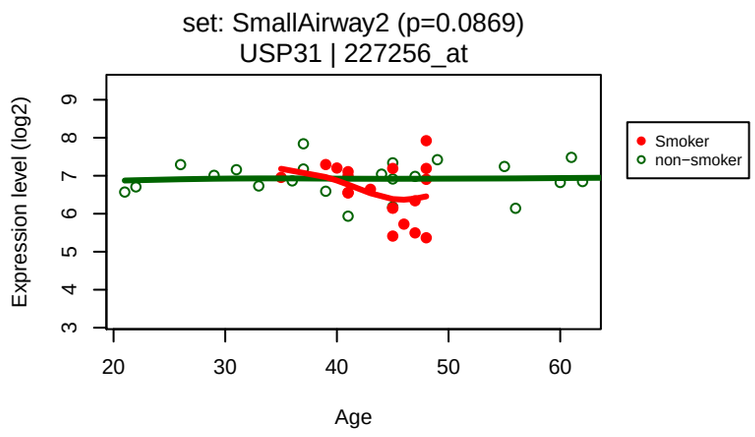
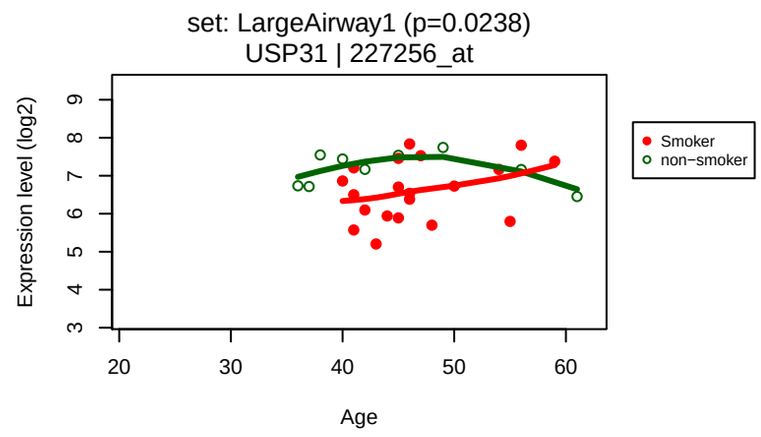

Gene expression of smokers and nonsmokers
gene: USP31 | 227256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1631
p=0.0869
p=0.193
p=0.0238
p=0.7591
p=0.1345
p=0.3467
p=0.1671
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

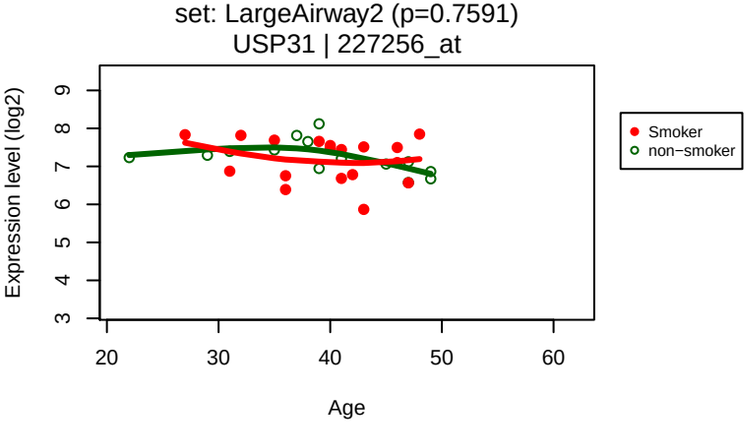

Gene expression ratio (smokers/nonsmokers)
gene: USP31 | 227256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1631
p=0.0869
p=0.193
p=0.0238
p=0.7591
p=0.1345
p=0.3467
p=0.1671
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

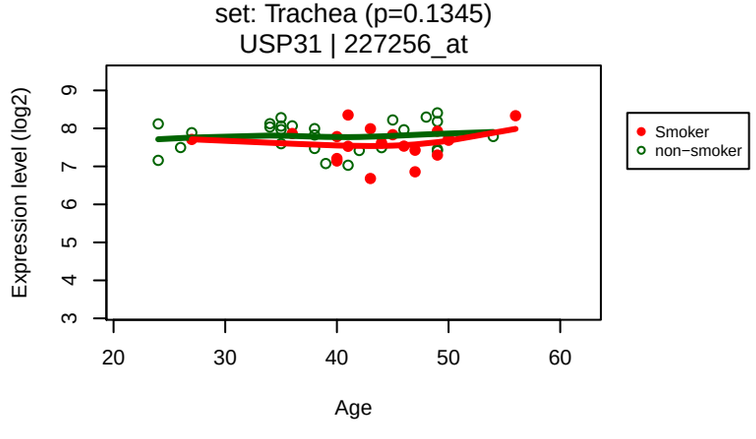

Gene expression ratio in Males
gene: USP31 | 227256_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1962
p=0.6852
p=0.2088
p=0.1728
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP31 | 227256_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2729
p=0.5752
p=0.7335
p=0.7366
SmallAirway2
LargeAirway2
Trachea
Macrophage2

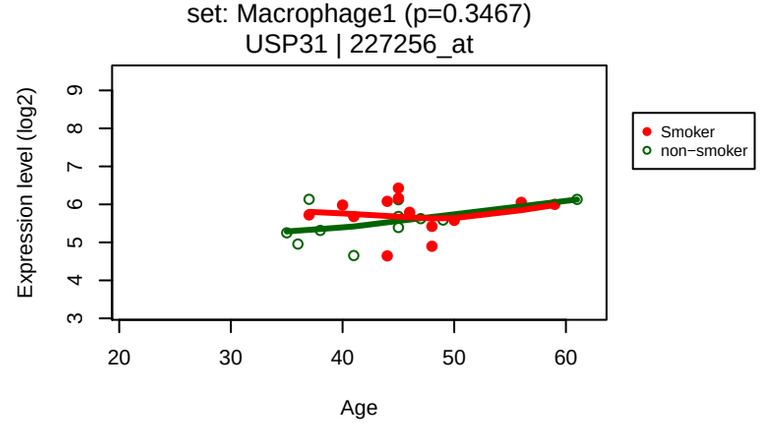

set: SmallAirway1 (p=0.1631)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker

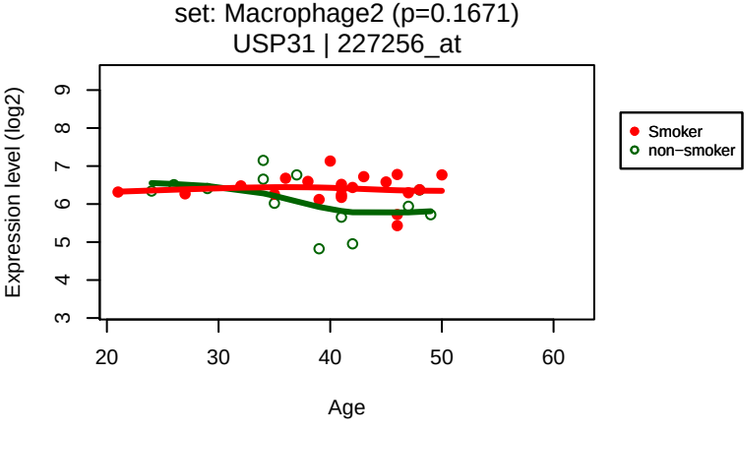

set: SmallAirway2 (p=0.0869)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker

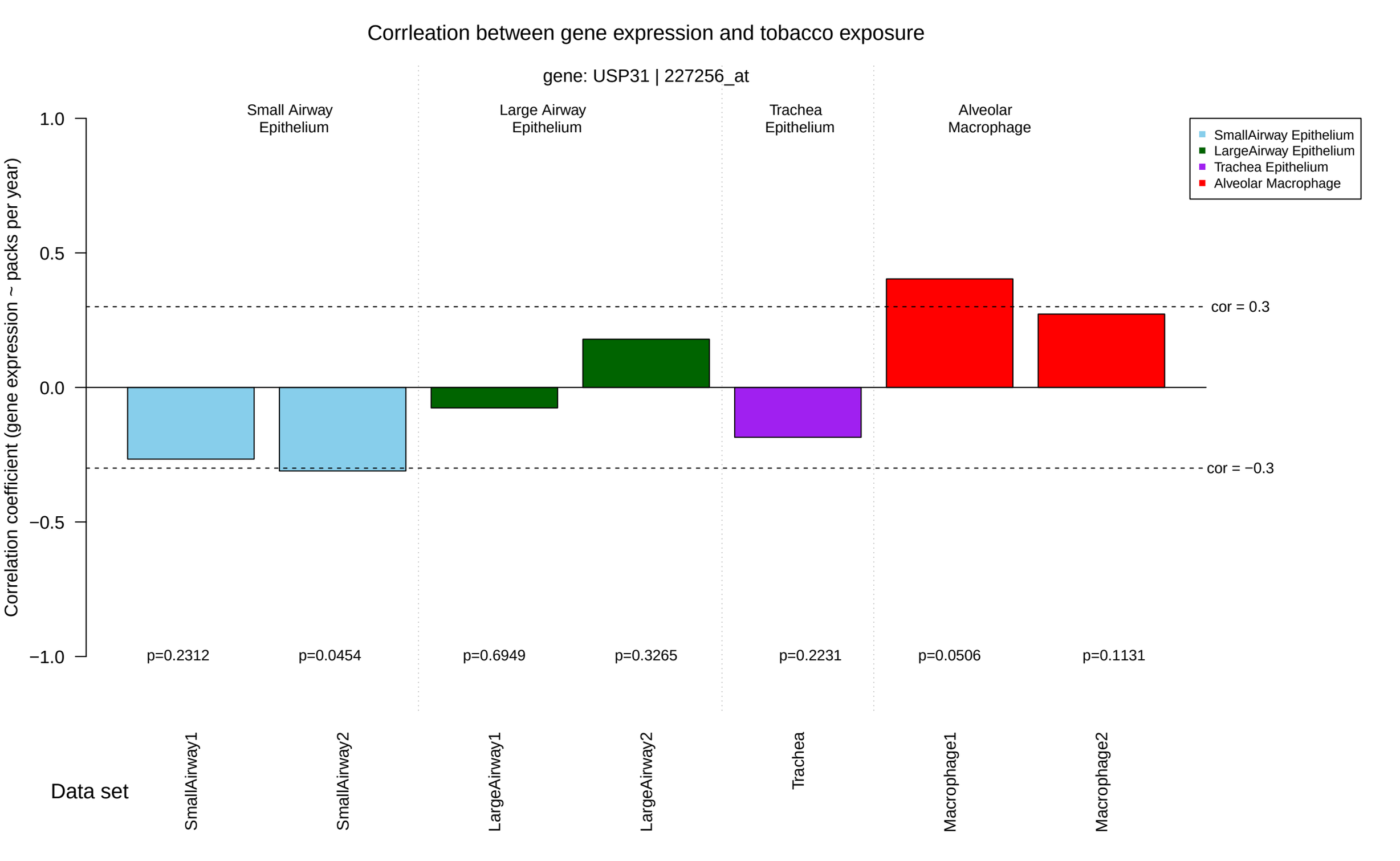

set: LargeAirway1 (p=0.0238)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7591)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1345)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3467)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1671)
USP31 | 227256_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP31 | 227256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2312
p=0.0454
p=0.6949
p=0.3265
p=0.2231
p=0.0506
p=0.1131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2